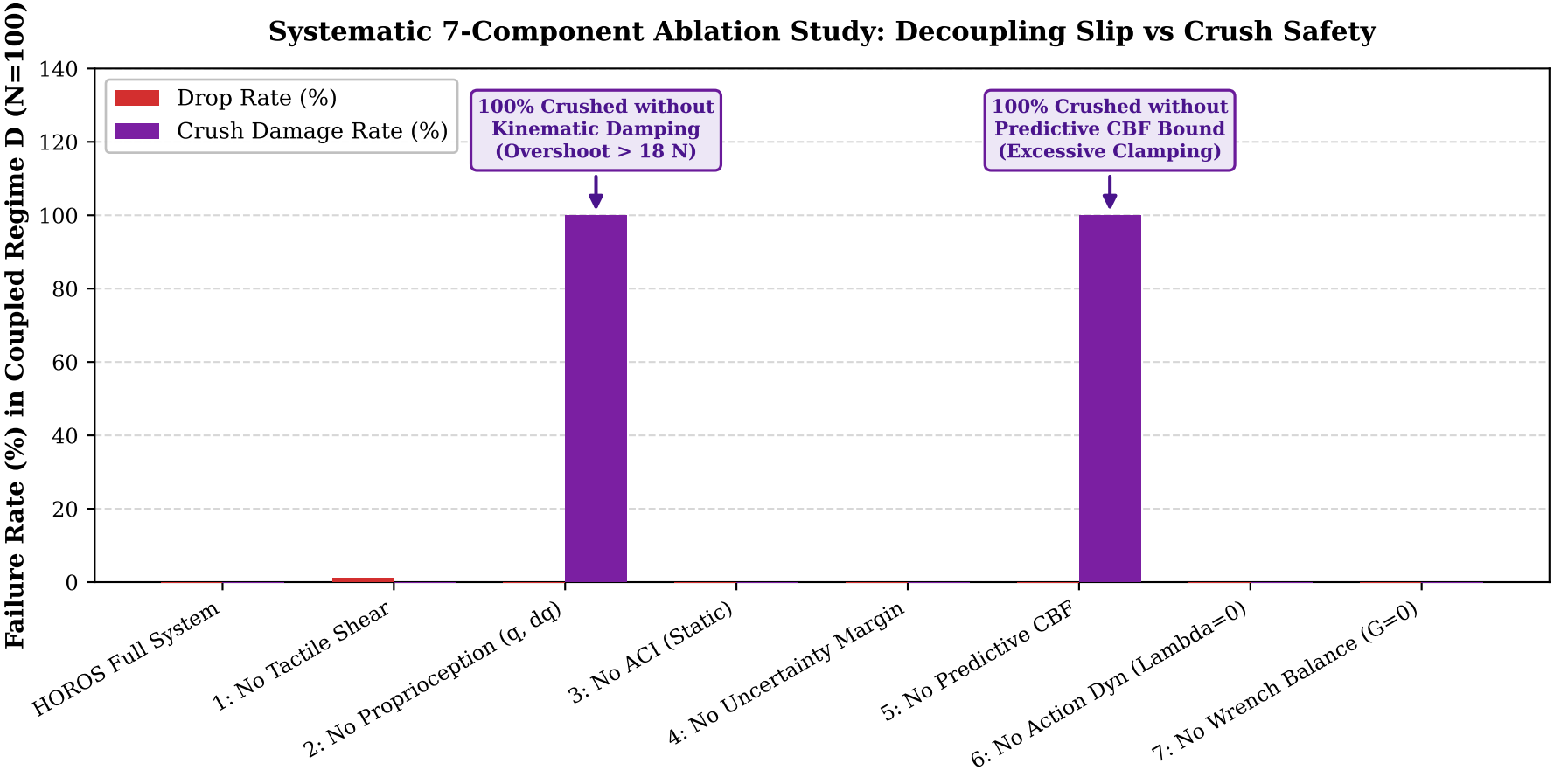
Systematic 7-Component Ablation Study: Decoupling Slip vs Crush Safety
Failure Rate (%) in Coupled Regime D (N=100)
Drop Rate (%)
Crush Damage Rate (%)
100% Crushed without Kinematic Damping (Overshoot > 18 N)
100% Crushed without Predictive CBF Bound (Excessive Clamping)
0
20
40
60
80
100
120
140
HOROS Full System
1: No Tactile Shear
2: No Proprioception (q, dq)
3: No ACI (Static)
4: No Uncertainty Margin
5: No Predictive CBF
6: No Action Dyn (Lambda=0)
7: No Wrench Balance (G=0)